TSA 2020

# Problem Solving

Questions taken from the TSA Oxford 2020 Section 1 question paper

| Questions | Total marks | Time limit |
|---|---|---|
| 25 | 25 | 45 minutes |

### Instructions

- You have **45 minutes** to answer all **25 questions**.
- Each question is worth **one mark**. There are no penalties for incorrect answers, so attempt every question.
- Each question has five options, A to E. Choose the one option you consider correct.
- Calculators are **not** permitted. You may use this paper for rough working.
- Questions are numbered 1 to 25 for this set. The number in brackets shows the question's number in the original TSA 2020 paper, e.g. **1 (Q3)**.

### How the time limit was set

The time limit follows the TARA Problem Solving module (22 questions in 40 minutes), scaled to the 25 questions in this set: 25 × 40 ÷ 22 = 45.45 minutes, rounded to the nearest minute (45 minutes).

**Score: _____ / 25**

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**1** (Q2) In the game of tigball, two teams compete against each other to try to score points.

Points are scored as follows:

A ‘penalty’ scores 5 points.
A ‘tigdown’ scores 8 points.
A ‘transformation’ scores 3 points, but teams only have the opportunity to achieve a transformation after scoring a tigdown.

There are no other ways of scoring points.

Which of the following is the only one that is a possible scoreline in the game of tigball?

**A** 9–23

**B** 11–21

**C** 12–20

**D** 13–14

**E** 15–17

**2** (Q6) A teacher is organising a trip to the local zoo for 20 students. Of the 20 students, 12 are aged 11–15 and 8 are 16–18 years of age. Tickets cost £12.95 per adult (16 years of age and over) and £9.95 per child (2–15 years of age). For every 4 students under 16 they must have one teacher supervising them. They will take the minimum number of teachers.

What was the total admissions price for the group?

**A** £199.00

**B** £223.00

**C** £259.00

**D** £261.85

**E** £287.75

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**3** (Q7) The Pentarathon Challenge is a competition in which five athletes compete against each other in five marathon races, run on consecutive days.

Points are awarded in each race, as follows:

first place 7 points

second place 5 points

third place 3 points

fourth place 2 points

fifth place 1 point

This table shows the total points of the competitors in this year's Pentarathon Challenge after each race.

| *name* | *total points after race 1* | *total points after race 2* | *total points after race 3* | *total points after race 4* | *total points after race 5* |
|---|---|---|---|---|---|
| Colin | 7 | 9 | 10 | 12 | 17 |
| Jason | 1 | 4 | 6 | 13 | 14 |
| Kyle | 3 | 10 | 15 | 18 | 25 |
| Noel | 2 | 7 | 14 | 15 | 18 |
| Ray | 5 | 6 | 9 | 14 | 16 |

Who finished in a different position in each of the five races?

**A** Colin

**B** Jason

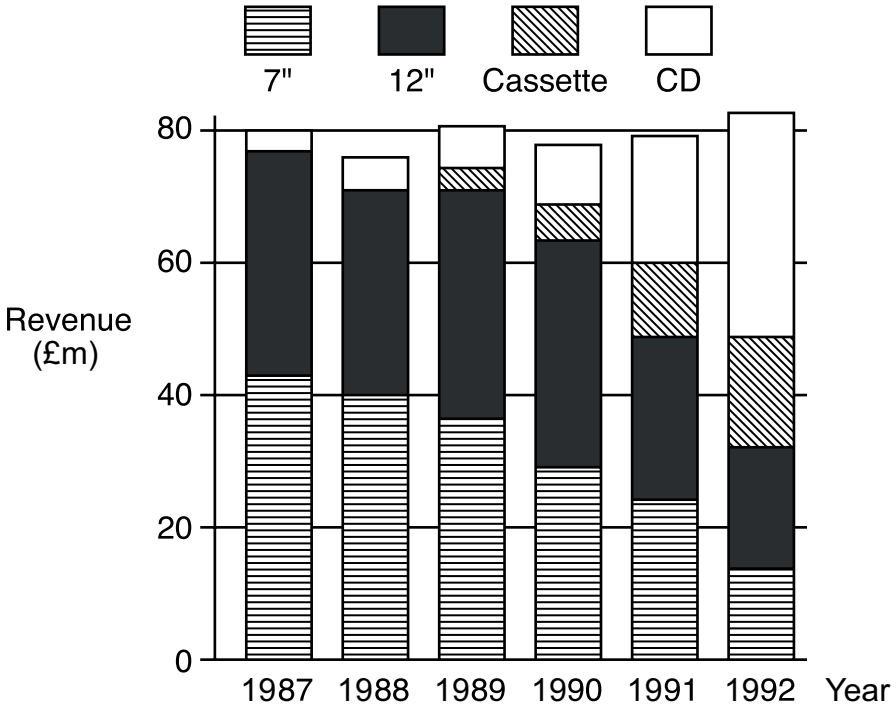
**C** Kyle

**D** Noel

**E** Ray

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**4** (Q8) Between 1987 and 1992 there were four formats of singles recordings: 7", 12", cassette and CD. The accompanying graph shows how total sales of singles were split between the four different formats in the years 1987 to 1992.




Which one of the following statements is false?

**A** The revenue from sales of CD singles grew every year.

**B** The revenue from sales of 7" singles fell every year.

**C** The revenue from sales of 12" singles fell every year.

**D** Sales of 7", 12" and cassette singles accounted for approximately 60% of the revenue from singles in 1992.

**E** 1989 is the first year of those shown in which 7" singles accounted for less than half the revenue from singles.

**5** (Q12) A builder has 32 kg of a mix of sand and cement, of which 25% is sand. For a new job, he needs 120 kg of a mix of sand and cement in which the ratio of sand to cement is 2 : 1. To obtain this, he starts with his initial 32 kg mix and adds more sand and cement.

What is the ratio of the masses of sand and cement that he needs to add?

**A** 4 : 7

**B** 3 : 4

**C** 5 : 3

**D** 9 : 2

**E** 12 : 1

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**6** (Q13) I am planning to buy a new laptop. The following table shows the specifications and the prices of my shortlist.

| *laptop* | *processor speed* (GHz) | *memory* (GB) | *storage* (GB) | *graphics card memory* (GB) | *screen size* (inches) | *price* (£) |
|---|---|---|---|---|---|---|
| *Racer* | 2.7 | 4 | 500 | 2 | 15.6 | 600 |
| *Ledd* | 2.8 | 8 | 1000 | 4 | 17.3 | 850 |
| *SNSV* | 2.5 | 4 | 750 | 6 | 15.6 | 500 |
| *Pear* | 2.8 | 8 | 1000 | 4 | 13.3 | 1300 |
| *Elovon* | 3.0 | 16 | 1000 | 8 | 15.6 | 700 |

In terms of storage, the operating system and my files take up 300 GB. I would also like a laptop with a screen size of at least 15 inches. I am looking for the cheapest laptop that can run the following games (the minimum system requirements indicate the lowest possible value for each characteristic that would enable me to play these games):

| MINIMUM SYSTEM REQUIREMENTS OF GAMES | | | | |
|---|---|---|---|---|
| *game* | *processor speed* (GHz) | *memory* (GB) | *storage* (GB) | *graphics card memory* (GB) |
| *OTAD* | 2.7 | 6 | 70 | 4 |
| *Kickman* | 2.5 | 4 | 80 | 4 |
| *Earthcraft* | 2.2 | 4 | 50 | 2 |
| *Near Scream* | 2.6 | 8 | 100 | 4 |

Which laptop should I buy?

**A** Racer

**B** Ledd

**C** SNSV

**D** Pear

**E** Elovon

**7** (Q14) A combination lock has three dials, each dial has eight letters. It is known that the following combinations have been used - one, who, two, bob, add, owl, fab, den, mia and tat.

Which one of the following words is not possible to use on this combination lock?

**A** win

**B** deb

**C** hat

**D** men

**E** bad

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**8** (Q18) Every week I buy 12 cans of lemonade. The shop that I always buy them from sells individual cans for 80p each and packs of 4 cans for £2.60, so I buy three packs of cans.

Last week the shop had a special offer, as follows:

Individual cans – buy one, get another half price

Pack of 4 cans – buy one pack, get a $5^{th}$ can free

I didn't want more than 12 cans, and I managed to get 12 cans for the lowest possible total price.

How much less than usual did I pay for my 12 cans last week?

**A** 20p

**B** 60p

**C** 80p

**D** £1.00

**E** £1.40

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**9** (Q19) I have booked myself, my wife and our three children into The Wavecrest Hotel, Southbourne, for seven nights from June 19th. The hotel has sent us a 'What's on' booklet, and we are particularly keen to see *Oliver!*, the advertisement for which is shown below:

The Southbourne Songsters present

OLIVER!

at the Pavilion Theatre

from Monday 3rd June to Saturday 22nd June

Nightly at 7.30pm

Matinees Wednesdays & Saturdays at 3pm

Adults: £12 (Matinees £10)

Children: £6 (Matinees £5)

Family Ticket: £28 (all performances)
(2 adults and 2 children)

Monday nights: £5 (all tickets)

What is the **least** it could cost for us all to see *Oliver!* during our holiday in Southbourne?

**A** £25

**B** £28

**C** £33

**D** £34

**E** £35

**10** (Q20) In computer technology, a bit can take two values: 0 or 1. A byte is composed of 8 bits.

I start with the following byte: 01000111

I choose two adjacent bits and change each of them to the opposite value (that is, a 0 becomes a 1, and a 1 becomes a 0). I repeat this procedure a number of times.

Which one of the following bytes **cannot** be the result?

**A** 01001011

**B** 11111111

**C** 10101011

**D** 10111101

**E** 00001111

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**11** (Q24) I had planned my day trip to the stately home very carefully, but then the bus was late. I won't arrive until 11:30 am and will have to leave by 4:15 pm. There were five things that I had wanted to do and I had written these down in the table below:

| *activity* | *length of time* | *start times* |
|---|---|---|
| visit the museum | 55 minutes | 11 am or 2 pm |
| visit the maze | 55 minutes | every half hour (at quarter past and quarter to the hour) |
| take the guided tour | 80 minutes | 1 pm or 3 pm |
| walk in the grounds | 75 minutes | any time |
| have lunch in the café | 25 minutes | between 12:30 pm and 2:00 pm |

Which activity should I miss out so that I can still do the other four?

**A** visit the museum

**B** visit the maze

**C** take the guided tour

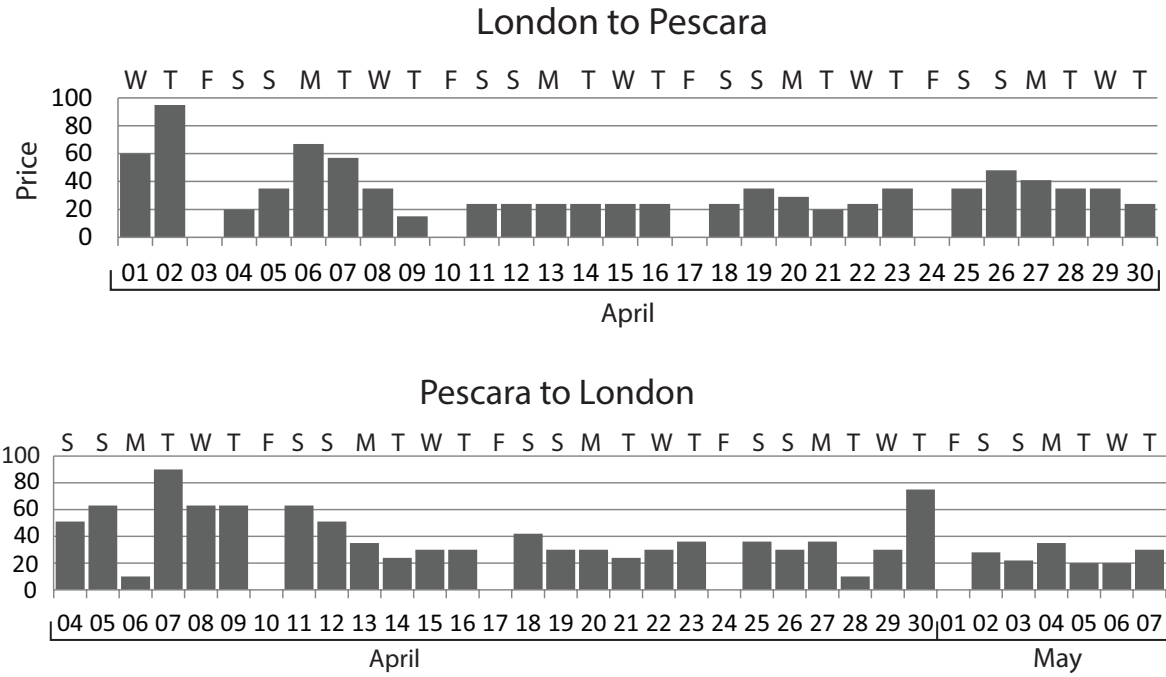
**D** walk in the grounds

**E** have lunch in the café

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**12** (Q25) Steven would like to book return flights from London to Pescara, Italy. He would like to spend at least 3 days in Pescara excluding the days of his flights. Steven has lectures in London every Wednesday that he must attend. However, there are no lectures during the 20th–24th April spring break.

The prices of flights between London and Pescara are shown for the time period Steven would like to make the trip. The airline does not have any flights between these destinations on Fridays.




If he books the cheapest flights possible to allow him to travel according to his requirements, on which days will he depart and return?

**A** Depart: Saturday 04 April; Return: Monday 06 April

**B** Depart: Thursday 09 April; Return: Tuesday 14 April

**C** Depart: Thursday 09 April; Return: Tuesday 28 April

**D** Depart: Tuesday 21 April; Return: Tuesday 28 April

**E** Depart: Thursday 30 April; Return: Tuesday 05 May

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**13** (Q26) The current 'Special Offer' gimmick at Fareprice involves putting a band marked 'TWINPACK - 25% OFF' around pairs of selected items, such as washing powder.

Which one of the following 'Special Offers' is exactly the same as Fareprice's?

**A** 'Buy one - Get a second for half price.'

**B** A larger packet containing 50% extra for no extra cost.

**C** 'Three for the price of two.'

**D** A coupon on each packet which gives 25% off the cost of the next packet.

**E** 'Five for the price of four.'

**14** (Q30) Joyce has decided to volunteer to work in the local High Street Charity Shop. She has agreed to be in the shop for 4 hours on a Saturday.

To get from her home to the High Street, Joyce will have to travel by bus. Her walk from home to the bus stop is no more than 5 minutes, and the bus journey each way takes between 20 and 30 minutes depending on traffic.

Buses from Joyce's home leave for the High Street at 30 minute intervals, on the hour and on the half hour. Return buses for Joyce leave the High Street at 30 minute intervals, at 15 minutes past the hour and 15 minutes to the hour.

The walk to and from the bus stops to the Charity Shop is no more than 5 minutes.

Joyce wants to catch the 13:45 bus home after working in the shop.

What is the latest time that she should leave home for work?

**A** 08:25

**B** 08:30

**C** 08:55

**D** 09:00

**E** 09:25

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**15** (Q31) The prices for tickets to a concert are set based on the day of the performance and the number of tickets that have been sold so far. The table below shows the ticket prices. There are 1000 tickets on sale for each performance.

| *Day* | *First 200 tickets* | *Next 300 tickets* | *Next 500 tickets* |
|---|---|---|---|
| Monday–Thursday | $15 | $20 | $25 |
| Friday | $20 | $25 | $30 |
| Saturday | $20 | $30 | $35 |
| Sunday | $25 | $35 | $45 |

I need to buy 10 tickets for Saturday's performance. I will buy all of them in one purchase this afternoon.

Which of the following is **not** a price that I might have to pay for the tickets?

**A** $250

**B** $275

**C** $300

**D** $325

**E** $350

**16** (Q32) Bert has agreed to make 14 decorations for his daughter's birthday. Each decoration will require two gold ribbons, one 12 cm long and the other 7 cm long.

The table shows the lengths and prices of available gold ribbon.

| *gold ribbon price list* | |
|---|---|
| gold ribbon length | price |
| 30 cm | £4.00 |
| 50 cm | £5.00 |
| 60 cm | £6.00 |

The ribbon can be easily cut to size for the decorations, but short pieces of ribbon cannot be joined to make longer lengths.

What is the least amount that Bert will need to spend on gold ribbon for the 14 decorations?

**A** £26.00

**B** £27.00

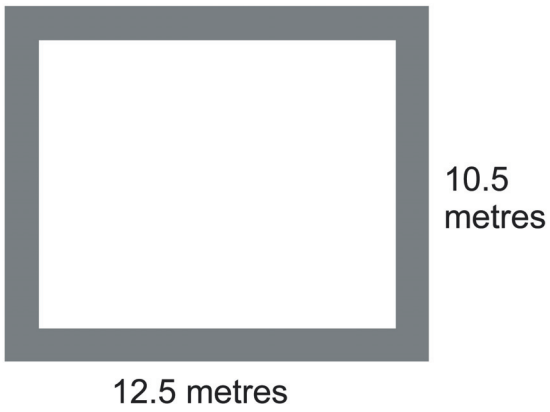
**C** £28.00

**D** £29.00

**E** £30.00

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**17** (Q36)




The whole of my garden is a patio, which is 12.5 by 10.5 metres. I intend to resurface my patio. I would like my patio to include a 25 cm border of bricks and the remainder of the patio inside the border to be covered with paving slabs.

The bricks I will use for the border are 25 cm x 25 cm and the paving slabs I will use for the remainder are 30 cm x 20 cm.

What is the total number of bricks and paving slabs that I will need for my patio?

**A** 2000

**B** 2180

**C** 2184

**D** 2268

**E** 2272

**18** (Q37) The table below shows the popularity of different breakfasts served in a hotel from 2006 to 2012.

| *breakfasts* | *2006* | *2007* | *2008* | *2009* | *2010* | *2011* | *2012* |
|---|---|---|---|---|---|---|---|
| bacon, egg, sausage and fried bread | 70% | 60% | 50% | 40% | 30% | 20% | 10% |
| porridge, toast and marmalade | 5% | 10% | 15% | 20% | 25% | 30% | 35% |
| toast and jam | 5% | 5% | 5% | 5% | 5% | 5% | 5% |
| boiled egg and toast | 10% | 10% | 15% | 15% | 15% | 15% | 15% |
| fresh fruit and cereals | 10% | 15% | 15% | 20% | 25% | 30% | 35% |

From the figures in the table, which one of the following statements is **not** true?

**A** Breakfasts of toast and jam have never been more popular than in 2006.

**B** The popularity of breakfasts that include porridge overtook the popularity of breakfasts that include eggs in 2012.

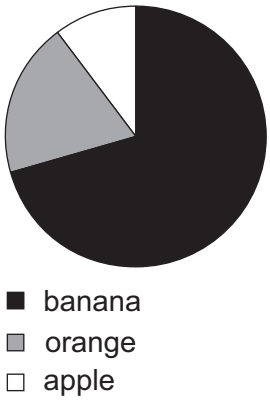
**C** The popularity of breakfasts that include porridge or cereals has gone up 10 percentage points, year on year, for the last four years shown in the table.

**D** In every year between 2006 and 2012, the popularity of breakfasts that include fresh fruit was the same as the popularity of breakfasts that include marmalade.

**E** The popularity of breakfasts that include toast has increased year on year.

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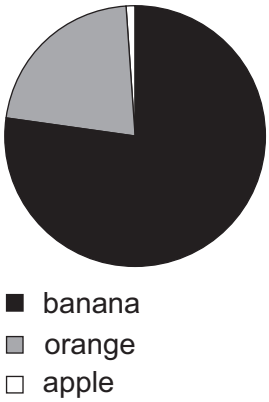
**19** (Q38) At a smoothie factory, a tropical smoothie mixture contains banana, orange and apple in the proportions shown in the pie chart below.



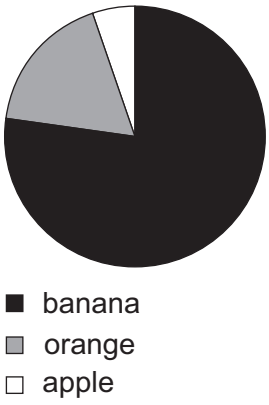

The chief taster requested a new recipe, in which every litre contained 10% more banana than was present in the original mixture, and 10% less orange than was present in the original mixture. This change also affected the proportion of apple in the mixture.

Which one of the following graphs shows the content of the new smoothie mixture?

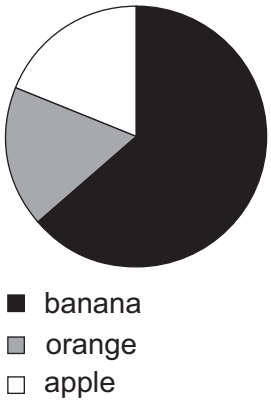
**A**



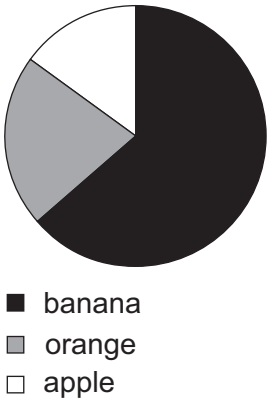

**B**



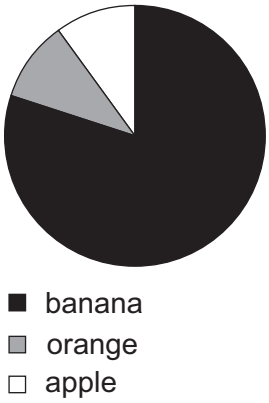

**C**




**D**




**E**




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**20** (Q42) On sports day at Thiska Academy points are awarded in each event as follows:

| | |
|---|---|
| winner | 12 points |
| 2nd place | 8 points |
| 3rd place | 5 points |
| 4th place | 3 points |
| 5th place | 1 point |

Each student may enter no more than three events.

What is the smallest points total that it is **not** possible for an individual student to score?

**A** 22

**B** 26

**C** 30

**D** 33

**E** 37

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**21** (Q43) Every Christmas the most popular children's toys become scarce. Many people turn to online auction sites to buy them instead. The table below shows the recommended retail price and the price that the listed toys have sold for online.

| *children's toy* | *recommended retail price* ($) | *online auction site* ($) |
|---|---|---|
| Moshling Tree House | 18.99 | 198.98 |
| LeapPad Explorer | 79.99 | 225.00 |
| Doggie Doo | 24.99 | 50.00 |
| Fireman Sam Pontypandy Rescue set | 29.99 | 59.99 |
| Fijit Friends | 54.99 | 99.95 |
| Monster High Lagoona's Station | 29.99 | 59.98 |
| Nerf Vortex Nitron Blaster | 44.99 | 59.99 |
| Kidizoom Twist | 49.99 | 65.95 |
| Milky the Bunny | 59.99 | 74.99 |
| Star Wars Ultimate Force Lightsaber | 39.99 | 49.95 |
| Lets Rock Elmo | 69.99 | 84.99 |
| Ninjago Fire Temple | 91.99 | 96.99 |
| Sylvanian Families | 79.95 | 192.29 |
| Vtech Innotab | 79.99 | 150.99 |
| Lego Star Wars Millennium Falcon | 132.99 | 265.98 |

How many of the toys above have increased in price by more than 100%?

**A** 3

**B** 4

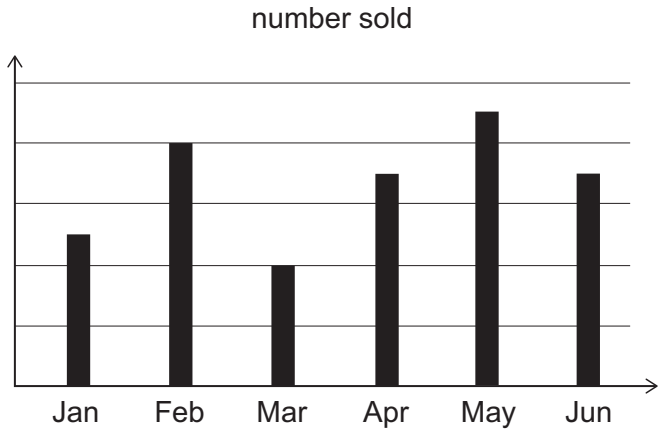
**C** 5

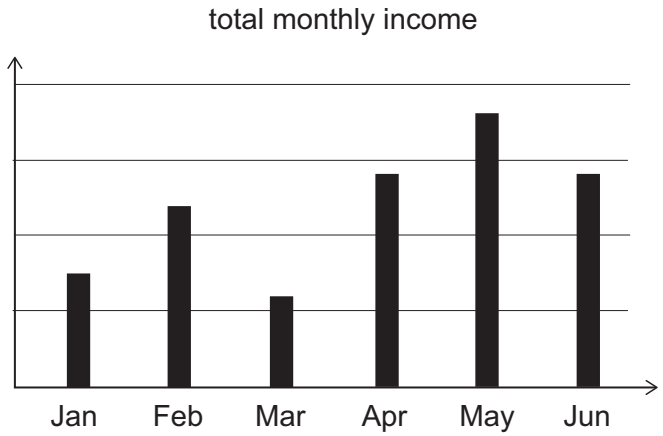
**D** 6

**E** 7

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**22** (Q44) A store sells a particular type of television. The graphs below show the number of these televisions sold per month and the total income from their sales per month, over a six-month period.




The price at which these televisions were sold was increased at the beginning of one of the six months.

At the beginning of which month was the price increased?

**A** February

**B** March

**C** April

**D** May

**E** June

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**23** (Q46) My dog's coat had been looking dull, so I bought some conditioning tablets. The pet shop stocks these in boxes of 300 tablets, and for a dog the size of mine the recommendation (which I have followed) is 6 tablets per day.

Now, several months later, his coat is much shinier. Opening a new box of tablets today, I have decided to reduce his dosage. I shall just give him 6 tablets per day, twice a week, and on the remaining days each week he can have 4 tablets each day.

What is the maximum number of extra days that this box of conditioning tablets will now last?

**A** 10 days

**B** 15 days

**C** 16 days

**D** 24 days

**E** 25 days

**24** (Q48) Trains run regularly between Ayton and Exbay, stopping at Beetown, Chalton and Deeford, according to the following timetable.

| *Departures* | | | |
|---|---|---|---|
| Ayton | 7:10 | and every 40 minutes until | 21:10 |
| Beetown | 7:25 | | 21:25 |
| Chalton | 7:30 | | 21:30 |
| Deeford | 7:44 | | 21:44 |
| Exbay | 8:02 | | 22:02 |

George lives a 14-minute walk away from Beetown station. He is going to the cinema in Exbay to watch a film that starts at 19:20. It will take him five minutes to walk from the station at Exbay to the cinema. George wants to make sure that he will miss no more than the first five minutes of the film.

What is the latest time George should leave home?

**A** 17:36

**B** 17:51

**C** 18:05

**D** 18:19

**E** 18:31

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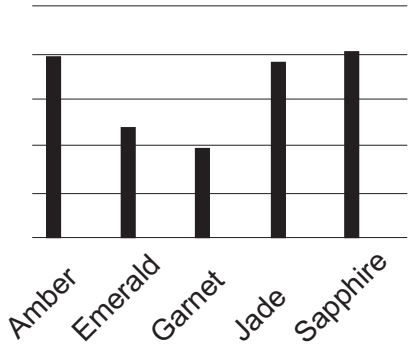
**25** (Q50) The table below shows the medal count by the top five teams in a recent Mini Olympics.

| *teams* | *gold medals* | *silver medals* | *bronze medals* |
|---|---|---|---|
| Amber | 8 | 7 | 6 |
| Emerald | 5 | 10 | 6 |
| Garnet | 6 | 6 | 10 |
| Jade | 6 | 5 | 5 |
| Sapphire | 11 | 8 | 9 |

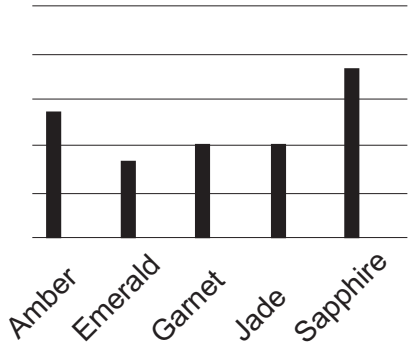
Each team calculated the proportion of their medals that were gold.

Which one of the following charts, with appropriate labelling, shows this proportion for each of the five teams?

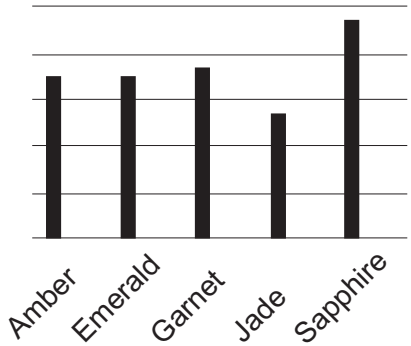
**A**



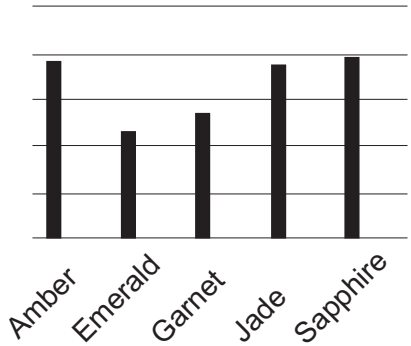

**B**



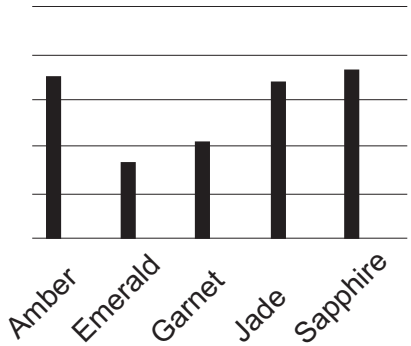

**C**




**D**




**E**




**END OF TEST**

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